

The lower asymptotic mode threshold for trees and forests is the same

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Abstract

Let m be any global mode of the independence sequence of an n -vertex forest F , and let $\mu(F)$ be the mean size of a uniformly random independent set. We give a self-contained bound $|m - \mu(F)| < 6n^{3/4}$. A one-hub construction then proves that the minimum normalized global mode, over trees or over forests of order exactly n , converges to the same constant. This constant is the infimum of $\mu(T)/\alpha(T)$ over finite nonempty trees. An alternating decorated path gives the upper bound $\frac{5}{8}(1 - 1/\sqrt{17})$. No unimodality theorem or local limit theorem is required for the main result.

1 Statement and scope

For a finite graph G , put

$$Z_G(x) = \sum_{k=0}^{\alpha(G)} i_k(G)x^k, \quad \mu(G) = \frac{Z'_G(1)}{Z_G(1)}, \quad \text{Mode}(G) = \operatorname{argmax}_k i_k(G).$$

All graphs below are simple. Forests and trees in infima are nonempty. For $\mathcal{C}$ equal to the class of trees or forests, define

$$A_{\mathcal{C}}(n) = \min_{\substack{G \in \mathcal{C} \\ |V(G)|=n}} \min_{m \in \text{Mode}(G)} \frac{m}{\alpha(G)}.$$

These minima exist because there are finitely many graphs of order n .

Theorem 1. *With*

$$a_* := \inf_{T \text{ finite tree}} \frac{\mu(T)}{\alpha(T)},$$

one has the limits over all positive integer orders

$$\lim_{n \rightarrow \infty} A_{\text{tree}}(n) = \lim_{n \rightarrow \infty} A_{\text{forest}}(n) = a_*. \quad (1)$$

Moreover,

$$a_* \leq \frac{5}{8} \left(1 - \frac{1}{\sqrt{17}} \right) = 0.473415234352 \dots \quad (2)$$

Every n -vertex forest satisfies, for every global mode m ,

$$\left| \frac{m}{\alpha(F)} - \frac{\mu(F)}{\alpha(F)} \right| < 12n^{-1/4}. \quad (3)$$

The exact value of a_* is not determined here; in particular, equality in (2) is not claimed. Fang et al. [1] ask in Section 9 whether the lower asymptotic peak threshold for trees equals that for forests. Equation (1) answers the global-mode version of this question. The relation to their definition by weak local peaks is spelled out in Section 5. This is an unreviewed proof manuscript, not a proof-assistant certificate. No assertion of publication priority is made.

2 An explicit variance bound without a degree restriction

Lemma 2. *For a uniformly random independent set I of an n -vertex forest,*

$$\text{Var } |I| \leq \frac{3 + 2\sqrt{2}}{2} n < 3n.$$

Proof. Write $S_v = \mathbf{1}_{\{v \in I\}}$, $p_v = \mathbb{E} S_v$, $s_v^2 = p_v(1 - p_v)$, and $w_v = p_v/(1 - p_v)$. All s_v are positive. For an oriented edge $u \rightarrow v$, let $R_{u \rightarrow v}$ be the occupation odds at u in the component containing u after deleting uv . The standard occupied/unoccupied partition of independent sets gives

$$R_{u \rightarrow v} = \prod_{z \sim u, z \neq v} (1 + R_{z \rightarrow u})^{-1} \in (0, 1], \quad w_v = \prod_{u \sim v} (1 + R_{u \rightarrow v})^{-1}.$$

Since adjacent vertices cannot both be occupied, their correlation c_{uv} satisfies

$$c_{uv} = -\sqrt{w_u w_v}, \quad c_{uv}^2 = w_u w_v.$$

Fix v and put $x_u = R_{u \rightarrow v}$. Since $w_u = x_u/(1 + R_{v \rightarrow u}) \leq x_u$, we obtain

$$\sum_{u \sim v} c_{uv}^2 \leq \frac{\sum_{u \sim v} x_u}{\prod_{u \sim v} (1 + x_u)} \leq \frac{1}{2}. \quad (4)$$

For completeness, the last inequality holds for any finite collection $x_i \in [0, 1]$. The difference $\prod_i (1 + x_i) - 2 \sum_i x_i$ is multiaffine, so its minimum on the cube is attained at a vertex. At a vertex with j entries equal to one, its value is $2^j - 2j \geq 0$. The empty collection also satisfies the inequality.

Root each component and order the vertices with every parent before its children. Set $Y_v = (S_v - p_v)/s_v$. Define a matrix B by

$$B_{v,u} = \begin{cases} c_{uv}, & u \text{ is the parent of } v, \\ 0, & \text{otherwise.} \end{cases}$$

Its columns have disjoint supports, so (4) implies $\|B\|_2^2 \leq 1/2$. Let $\epsilon = (I - B)Y$. For a root, $\epsilon_v = Y_v$; otherwise, $\epsilon_v = Y_v - c_{uv}Y_u$ for its parent u .

Conditional on the parent state, a descendant subtree is independent of the rest of the forest. Also a centered function of a binary parent is a scalar multiple of Y_u , and the scalar here is $\mathbb{E}(Y_v Y_u) = c_{uv}$. Thus

$$\mathbb{E}(\epsilon_v \mid \text{all previously revealed states}) = 0.$$

Consequently the innovations are pairwise orthogonal. Their covariance matrix D is diagonal, with root entries 1 and other entries $1 - c_{uv}^2$, hence $0 \preceq D \preceq I$. The matrix B is nilpotent, and

$$\text{Cov}(Y) = (I - B)^{-1} D (I - B)^{-\top}, \quad \|\text{Cov}(Y)\|_2 \leq (1 - 1/\sqrt{2})^{-2}.$$

Finally $|I| - \mathbb{E}|I| = \sum_v s_v Y_v$ and $\sum_v s_v^2 \leq n/4$. This proves the stated bound. $\square$

Lemma 3. *Every global mode m of the independence sequence of an n -vertex forest satisfies $|m - \mu(F)| < 6n^{3/4}$.*

Proof. Let $X = |I|$ and $\mu = \mathbb{E}X$. By Lemma 2 and Chebyshev's inequality, the interval $[\mu - 2\sqrt{3n}, \mu + 2\sqrt{3n}]$ has probability at least $3/4$. It contains at most $4\sqrt{3n} + 1 < 8\sqrt{n}$ integers. Therefore every global mode satisfies

$$\mathbb{P}(X = m) \geq \frac{3}{32\sqrt{n}}.$$

Using the single summand at m in the variance gives

$$\frac{3(m - \mu)^2}{32\sqrt{n}} \leq \text{Var } X < 3n, \quad |m - \mu| < \sqrt{32} n^{3/4} < 6n^{3/4}.$$

Every forest is bipartite, so $\alpha(F) \geq n/2$, proving (3) as well. $\square$

3 One hub transfers every finite mean ratio to trees

Lemma 4. *Fix a nonempty finite tree H and a vertex v of H . There are trees T_n of every order $n \geq |V(H)| + 1$ such that*

$$\frac{\mu(T_n)}{\alpha(T_n)} \longrightarrow \frac{\mu(H)}{\alpha(H)}.$$

The same limit holds if $\mu(T_n)$ is replaced by any choice of global mode of T_n .

Proof. Let $h = |V(H)|$ and write $n - 1 = rh + s$, $0 \leq s < h$. Take r disjoint copies of H , join a new hub to the copy of v in each, and attach s additional leaves to the hub. The resulting graph is a tree of exactly n vertices. Conditioning on the hub gives the exact identity

$$Z_{T_n}(x) = (1 + x)^s Z_H(x)^r + x Z_{H-v}(x)^r. \quad (5)$$

Put $\theta = Z_{H-v}(1)/Z_H(1)$. There are independent sets containing v and independent sets avoiding v , so $0 < \theta < 1$. The probability p that the hub is occupied is

$$p = \frac{\theta^r}{2^s + \theta^r} \leq \theta^r.$$

The conditional means, with the hub absent and present respectively, are $r\mu(H) + s/2$ and $1 + r\mu(H - v)$. Thus

$$|\mu(T_n) - r\mu(H) - s/2| \leq n\theta^r. \quad (6)$$

Taking degrees in (5), with no cancellation since coefficients are nonnegative, gives

$$\alpha(T_n) = \max\{r\alpha(H) + s, 1 + r\alpha(H - v)\},$$

and hence

$$r\alpha(H) + s \leq \alpha(T_n) \leq r\alpha(H) + s + 1.$$

Here s is bounded as n varies and $r \rightarrow \infty$; dividing proves the mean-ratio limit. Lemma 3 proves the assertion for every global mode. $\square$

Proof of (1). If the components of a nonempty forest F are $H_1, \dots, H_j$, independence under the uniform measure and additivity of α give

$$\frac{\mu(F)}{\alpha(F)} = \sum_{i=1}^j \frac{\alpha(H_i)}{\alpha(F)} \frac{\mu(H_i)}{\alpha(H_i)} \geq a_*.$$

Thus (3) implies, for every $n \geq 1$,

$$A_{\text{tree}}(n) \geq A_{\text{forest}}(n) \geq a_* - 12n^{-1/4}.$$

On the other hand, for every fixed finite tree H , Lemma 4 gives

$$\limsup_{n \rightarrow \infty} A_{\text{tree}}(n) \leq \frac{\mu(H)}{\alpha(H)}.$$

Taking the infimum over H proves both limits and their equality. Notice that the construction treats every integer order, rather than a subsequence. $\square$

4 An explicit upper bound

Let C_t be the path with spine vertices $1, \dots, 2t$, with one new leaf attached to each odd spine vertex. It has $3t$ vertices and $\alpha(C_t) = 2t$: all even spine vertices and all pendant leaves form an independent set of this size, while the t disjoint leaf edges prohibit a larger one.

For a spine vertex with $a \in \{0, 1\}$ attached leaves, use the transfer matrix

$$M_a(x) = \begin{pmatrix} (1+x)^a & x \\ (1+x)^a & 0 \end{pmatrix},$$

whose rows record the previous spine state and columns the current state (unoccupied, occupied). Then

$$Z_{C_t}(x) = (1, 0) A(x)^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad A(x) = M_1(x)M_0(x) = \begin{pmatrix} 1+2x & x+x^2 \\ 1+x & x+x^2 \end{pmatrix}.$$

Its trace and determinant are $T(x) = 1+3x+x^2$ and $D(x) = x^2(1+x)$. Its Perron eigenvalue $L(x)$ is simple near $x = 1$, where

$$L(1) = \frac{5 + \sqrt{17}}{2}.$$

Both entries of the boundary vectors needed for the leading eigenprojection give a positive leading coefficient. The second eigenvalue has strictly smaller modulus, uniformly on a sufficiently small real neighborhood of 1. The two-eigenvalue formula and its derivative therefore give

$$\mu(C_t) = t \frac{L'(1)}{L(1)} + O(1).$$

Differentiating $L^2 - TL + D = 0$ yields

$$\frac{L'(1)}{L(1)} = \frac{5 - 5/L(1)}{\sqrt{17}} = \frac{5}{4} \left(1 - \frac{1}{\sqrt{17}} \right).$$

Since $\alpha(C_t) = 2t$, taking $t \rightarrow \infty$ proves (2).

5 Relation to weak local peaks

The main theorem above uses global modes throughout and is independent of unimodality. Fang et al. [1] define a peak using the weak local inequalities $i_{k-1} \leq i_k \geq i_{k+1}$. Unimodality alone would not exclude a flat plateau below the global maximum. Their stronger conclusions do exclude such plateaus for sufficiently large forests, as follows.

Their initial extension estimate is $(k+1)i_{k+1} \geq (n-3k)i_k$, giving strict increase when $k < (n-1)/4$. Their terminal extension estimate is $(k+1)i_{k+1} \leq 2(\alpha-k)i_k$, giving strict decrease when $k > (2\alpha-1)/3$. Theorem 1.2 of that paper gives strict log-concavity for $n/5 \leq k \leq 17\alpha/25$. For all sufficiently large n , these ranges overlap, since $\alpha \geq n/2$. Any weak local peak must therefore lie in the strict log-concavity interval. Within that interval the consecutive ratios i_k/i_{k-1} strictly decrease; a local peak is consequently a global mode, using strict monotonicity outside. Thus (1) also gives equality for their asymptotic lower peak thresholds. This last identification uses their theorem; the global-mode result does not.

Verification status

All constants used for (3) are explicit and valid for every positive integer n . The limiting hub argument uses a fixed H and its explicit $0 < \theta < 1$, not an unspecified uniform approximation across varying H . The accompanying Python script checks the hub polynomial identity by independent enumeration on small instances, exact finite variance and mode bounds, and the decorated-path transfer recurrence. Those checks are supplementary finite audits, not a substitute for the proofs above and not a formal proof-assistant certificate. The exact common threshold and the extremal density of log-concavity failures remain undetermined here.

References

- [1] E. X. Fang, J. Lu, E. Nevo, Y. Yao, and H. Zheng, *Unimodality of Independence Polynomials for Sufficiently Large Forests*, arXiv:2609.20961v1 (September 17, 2026), Sections 8–9 and Theorem 1.2. <https://arxiv.org/abs/2609.20961>.